# PART B – TRAFFIC CONTROL

## Section 14: Traffic Control Layouts – Pavement Marking

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# PART B – TRAFFIC CONTROL

## Section 14: Traffic Control Layouts – Pavement Marking

| LEGEND | | | |
|---|---|---|---|
| | Flexible Drum | | Paint Truck |
| | Tubular Marker | | Escort Truck |
| | Cone | | Chaser Vehicle |
| | Sign | | Vehicle Mounted Rear Crash Attenuator |
| | Traffic Control Person | | |
| | Work Activity Area | | 360° Flashing Light |
| | Work Truck | | Portable Traffic Signal |
| | Shadow Vehicle | | Barricade and Fencing |
| | Shadow Vehicle #1 | | Flashing Arrow Board (FAB) |
| | Shadow Vehicle #2 | | Flashing Arrow Board (FAB) in caution mode |
| | Buffer Vehicle | | Dynamic Message Sign (DMS) |

**TABLE A – TAPER LENGTHS**

| Taper Types (m) | | Regular Posted Speed Limit (km/h) | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | ≤50 | 60 | 70 | 80 | 90 | 100 | 110 | 120 |
| Merge Taper Length | $L_M$ | 35 | 55 | 160 | 190 | 210 | 230 | 250 | 280 |
| Lane Shift Taper Length | $L_L$ | 30 | 50 | 80 | 100 | 110 | 120 | 130 | 140 |
| Downstream Taper Length | $L_D$ | 30 | 30 | 30 | 30 | 30 | 30 | 30 | 30 |
| TCP, Signal, and Shoulder Taper Length (min. 5 devices) | $L_S$ | 5 | 8 | 15 | 15 | 15 | 15 | 15 | 15 |
| Minimum Tangent Length between Tapers | $L_T$ | 30 | 60 | 160 | 190 | 210 | 230 | 250 | 280 |
| Run-In Length on Centreline | $L_R$ | 40 | 50 | 60 | 60 | 70 | 80 | 90 | 100 |

**TABLE B – DEVICE SPACING LENGTHS**

| Device Spacing (m) | | Regular Posted Speed Limit (km/h) | | | | | | | |
|---|---|---|---|---|---|---|---|---|---|
| | | ≤50 | 60 | 70 | 80 | 90 | 100 | 110 | 120 |
| Construction Sign Spacing | A | 40 | 60 | 80 | 100 | 150 | 150 | 200 | 200 |
| Buffer Space | B | 30 | 40 | 60 | 80 | 110 | 140 | 170 | 200 |
| Roll-Ahead Buffer Distance | R | 30 | 30 | 40 | 40 | 40 | 50 | 50 | 50 |
| Channelizing Device Spacing for Tapers | C | 10 | 10 | 15 | 15 | 15 | 15 | 15 | 15 |
| Channelizing Device Spacing on Curves and Tangents | D | 10 | 10 | 30 | 30 | 40 | 40 | 40 | 50 |

## 14.1 General Information - Pavement Marking

1. For the purposes of this Manual, pavement marking is considered to be either conventional or quick-dry pavement marking:
   - **Conventional Pavement Marking**: When the drying time is 5 minutes or more after the passing of the paint truck and shadow vehicle, and where vehicles could track paint onto the roadway upon contact with it, it is necessary to implement advance warning signage and protection of the painted area until the paint is set. This involves using signs, cones, barricades, closures, and/or other traffic control devices.
   - **Quick-Dry Pavement Marking:** Quick-dry painting is a continuously-moving operation and differs from conventional longitudinal line painting by using dynamic message signs (DMS) on shadow vehicles to inform drivers that painting is in progress along the roadway. When the drying time is 90 seconds or less after the passing of the paint truck and shadow vehicle, the paint is set to a point where vehicles will not track paint onto the roadway upon contact with it.
2. Note that these two terms refer to the set time of the paint, not to two different applications. Paint that sets more slowly can typically be protected until it dries, while paint that sets more quickly may not require the same protection. For traffic control purposes, the determining factor is whether or not the paint will be tracked onto the roadway if vehicles come into contact with the freshly painted line.
3. The set time of the paint is determined by the composition of the paint and the condition and temperature of the surface and the air. Some paints that dry quickly under ideal conditions may no longer be “quick-dry” paints when used in less than ideal conditions. If the paint’s set time is prolonged for any reason, it can be treated as “conventional” paint that requires advance warning signage and protection of the freshly painted surface.

The traffic control layouts are considered the minimum standard. The associated text description of each figure highlights the key standards as well as guidance and options that can be considered by the user. The following information is typical on most layouts in this section:

**Standard:**

- All static signs and dynamic message signs (DMS) shall be visible to drivers when painting is in progress.
- All work vehicles require **two** 360-degree flashing lights when painting the centreline or working at night.
- Constant communication is required between all Vehicle Operators.
- Stationary construction signs and cones are not required for quick-dry marking.
- **Escort Vehicle Requirements:**
  - 360-degree flashing light and 4-way flashers
  - caution pattern or arrow on sequential arrow board
  - dynamic message sign (DMS)
  - Slow Moving Vehicle C-036 sign
  - Caution – Paint Spray Truck Ahead C-043 sign
  - Pass This Side C-042-L sign
- **Paint Truck Requirements:**
  - 360-degree flashing light and 4-way flashers
  - caution pattern or arrow on sequential arrow board
  - Wet Paint C-037-1 sign
  - Slow Moving Vehicle C-036 sign
  - Wet Paint – Keep Off C-037-2 sign
- **Shadow Vehicle #1 Requirements:**
  - 360-degree flashing light and 4-way flashers
  - caution pattern or arrow on sequential arrow board
  - dynamic message sign (DMS)
  - Slow Moving Vehicle C-036 sign
  - Wet Paint – Keep Off C-037-2 sign
  - truck-mounted crash attenuator for speeds ≥ 70 km/h
- **Shadow Vehicle #2 Requirements:**
  - 360-degree flashing light and 4-way flashers
  - caution pattern or arrow on sequential arrow board
  - dynamic message sign (DMS)
  - Slow Moving Vehicle C-036 sign

**Guidance:**

- None at this time.

**Options:**

- A crash attenuator may be added to Shadow Vehicle #2.

## 14.2 Quick-Dry Long-Line Marking – Two-Lane, Two-Way Roadway – Short and Long Duration

**Purpose:**

This layout shows the typical setup for quick-dry long-line marking on a two-lane, two-way roadway.

**Standard:**

**<u>Two-Lane, Two-Way Roadway – Passing NOT Permitted</u>:**

**Escort Operations:**

- The Escort Vehicle is positioned 200 to 500 meters in front of the Paint Truck, depending on traffic volumes and sight conditions.
- The DMS display shall be visible to oncoming traffic. It may be:
  - caution pattern
  - Line Painting … Ahead
  - Line Painting … Slow Down
- The Escort Vehicle Operator shall have a STOP/SLOW TCP C-027 paddle for emergencies.

**Shadow Vehicle #1 Operations:**

- Shadow Vehicle #1 follows the Paint Truck at a distance of 200 to 400 metres, depending on sight distances and paint drying times.
- The DMS shall display one these messages:
  - Line Painting … No Passing
  - Wet Paint … No Passing

**Shadow Vehicle #2 Operations:**

- Shadow Vehicle #2 follows the paint operation at a distance of approximately 1 kilometre, depending on sight distances, travelling on the shoulder where possible.
- The DMS messaging is specified by the foreman, and may be:
  - caution pattern
  - Line Painting … Ahead
  - Line Painting … 1 km Ahead
  - Line Painting … Next 1 km
  - Wet Paint … Next 1 km

**Two-Lane, Two-Way Roadway – Controlled Passing Opportunities:**

**Escort Vehicle Operations:**

- The Escort Vehicle should be stopped as close as possible to the centreline—but not straddling it—and have good visibility to oncoming traffic.
- When controlled passing is not occurring, the Escort Vehicle Operator shall display these messages on the DMS:
  - Line Painting … Ahead
  - Line Painting … Slow Down
- Upon notification, the Escort Vehicle Operator shall evaluate sight distances and traffic conditions before initiating controlled passing.
- If conditions are satisfactory for controlled passing, the Escort Vehicle Operator shall display STOP on the DMS.
- The Escort Vehicle Operator shall ensure that all vehicles have complied with the STOP message before allowing the paint operation to be passed.
- The Escort Vehicle Operator shall continue to hold stopped traffic until the last vehicle in the passing procession (as described by Shadow Vehicle #1 Operator) has cleared the work zone.
- The Escort Vehicle Operator shall notify the Shadow Vehicle #1 Operator that passing is complete.

**Shadow Vehicle #1 Operations:**

- When passing is required, the Shadow Vehicle #1 Operator shall communicate a passing request to the Escort Vehicle Operator and wait for notification that oncoming traffic is stopped.
- When it is safe to pass, the DMS display shall be changed to:
  - Line Painting … Pass on Left
  - Wet Paint … Pass on Left
- The Shadow Vehicle #1 Operator shall provide a description of the last vehicle in the passing procession to the Escort Vehicle Operator.
- Once the last vehicle has passed, the DMS display shall be changed to:
  - Line Painting … No Passing
  - Wet Paint … No Passing

**Shadow Vehicle #2 Operations:**

- On a two-lane roadway, Shadow Vehicle #2 assumes the buffer position.
- On a highway with a 3-metre shoulder, Shadow Vehicle #2 shall be driven along the right shoulder—not straddling the edge line—and follow the Paint Truck at a distance of approximately 1 kilometre.
- If the shoulder is too narrow, Shadow Vehicle #2 shall travel leap-frog fashion from approach to approach, parking parallel to the highway when stopping, keeping the signs and DMS visible to approaching traffic, and maintaining the approximate 1 kilometre distance behind the Paint Truck.
- The DMS messaging is specified by the foreman, and may be:
  - caution pattern
  - Line Painting … Ahead/Next 1 km
  - Wet Painting … 1 km Ahead/ Next 1 km

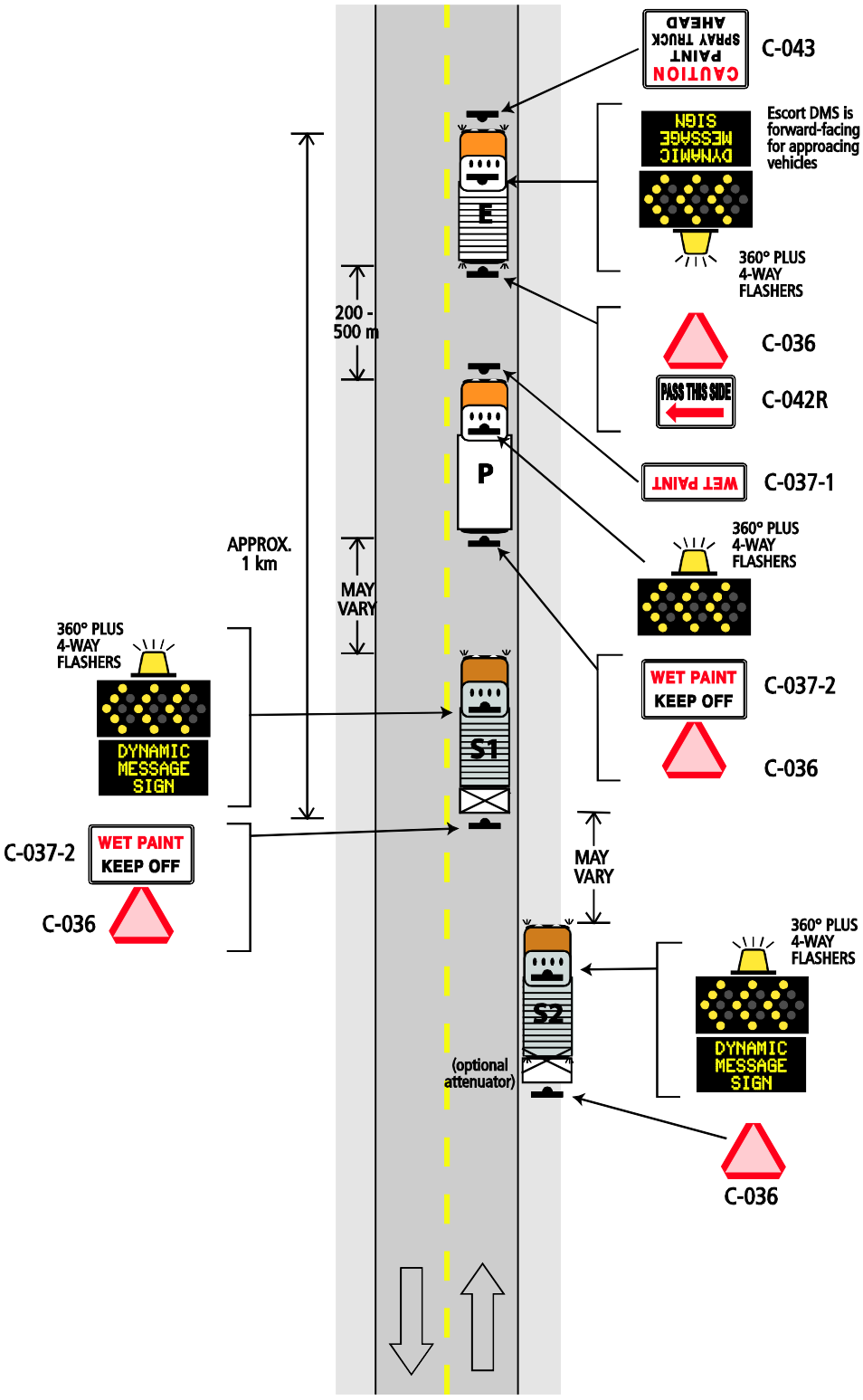
**Guidance:**

- The separation distance between the Paint Truck and Shadow Vehicle #1 should be determined by the set time (track-free time) of the pavement-marking paint.
- Sight distances and traffic volumes should be taken into consideration when choosing areas for controlled passing.
- Controlled passing of the painting operation requires effective communication and coordinated traffic control between the Escort Vehicle and Shadow Vehicle #1 Operators.
- Ideally, traffic will pass Shadow Vehicle #1 and the Paint Truck at the same time.
- When traffic volumes are excessive and the passing procession is large, the Paint Truck may have to stop the painting operation until the procession has passed completely. This will also help to reduce the required length of the passing zone.
- When a wide load or emergency vehicle is approaching the painting operation, all Vehicle Operators should be alerted so that they may take appropriate action.

**Options:**

- None at this time.

**Figure 14.2: Quick-Dry Long-Line Marking – Two-Lane, Two-Way Roadway – Short and Long Duration**



## 14.3 Quick-Dry Long-Line Marking – Right Lane – Multilane Roadway – Short and Long Duration

**Purpose:**

This layout shows the typical setup for quick-dry long-line marking on the right lane of a divided or undivided multilane roadway.

**Standard:**

**Shadow Vehicle #1 Operations:**

- Shadow Vehicle #1 shall travel in the same lane as the Paint Truck, following at a distance of 50 to 200 metres, depending on sight distance and traffic volumes. (It may have to be closer to the Paint Truck to prevent vehicles from entering the gap between the two vehicles, with consideration given to the amount of set time required for the paint.)
- The DMS messaging may be:
  - left chevron
  - Wet Paint … Keep Left
  - Line Painting … Keep Left

**Shadow Vehicle #2 Operations:**

- Shadow Vehicle #2 shall travel on the right shoulder—not straddling the edge line—and follow the Paint Truck at a distance of approximately 1 kilometre.
- The DMS messaging is specified by the foreman, and may be:
  - left chevron
  - Line Painting … Keep Left
  - Line Painting … Merge Left

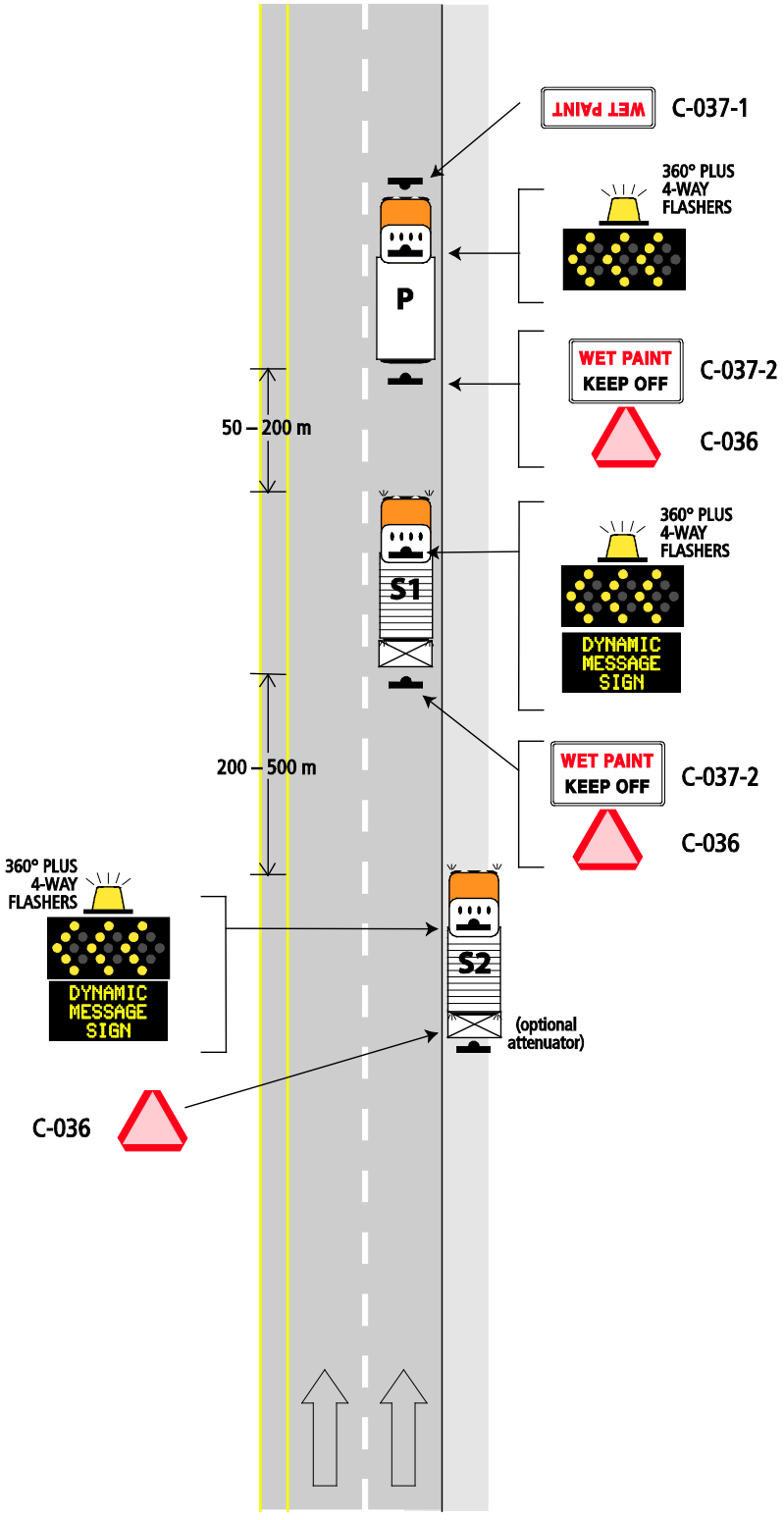
**Guidance:**

- The distance between the Paint Truck and Shadow Vehicle #1 should be determined by the set time (track-free time) of the pavement-marking paint.

**Options:**

- When traffic volumes are high, a third shadow vehicle may be positioned between the Paint Truck and Shadow Vehicle #1. It should follow the Paint Truck at a distance of 50 to 200 metres, followed by Shadow Vehicle #1 at a distance of 200 to 300 metres. The DMS messaging on the third shadow vehicle may be:
  - Wet Paint … Keep Left
  - Line Painting … Keep Left
- The third shadow vehicle may be the Escort Vehicle with the DMS repositioned to be rear-facing so that the message is visible to rear-approaching traffic.

**Figure 14.3: Quick-Dry Long-Line Marking – Right Lane – Multilane Roadway – Short and Long Duration**



## 14.4 Quick-Dry Long-Line Marking – Left Lane – Multilane Roadway (Median ≥ 1 m) – Short and Long Duration

**Purpose:**

This layout shows the typical setup for quick-dry long-line marking on the left lane of a multilane roadway when the centre median is ≥ 1 metre wide.

**Standard:**

**Shadow Vehicle #1 Operations:**

- Shadow Vehicle #1 shall travel in the same lane as the Paint Truck, following at a distance of 50 to 200 metres, depending on sight distance and traffic volumes. (It may have to be closer to the Paint Truck to prevent vehicles from entering the gap between the two vehicles, with consideration given to the amount of set time required for the paint.)
- The DMS messaging may be:
  - right chevron
  - Wet Paint … Keep Right
  - Line Painting … Keep Right

**Shadow Vehicle #2 Operations:**

- Shadow Vehicle #2 shall travel as far left as practicable, following the Paint Truck at a distance of approximately 1 kilometre.
- The DMS messaging is specified by the foreman, and may be:
  - right chevron
  - Line Painting … Keep Right
  - Line Painting … Merge Right

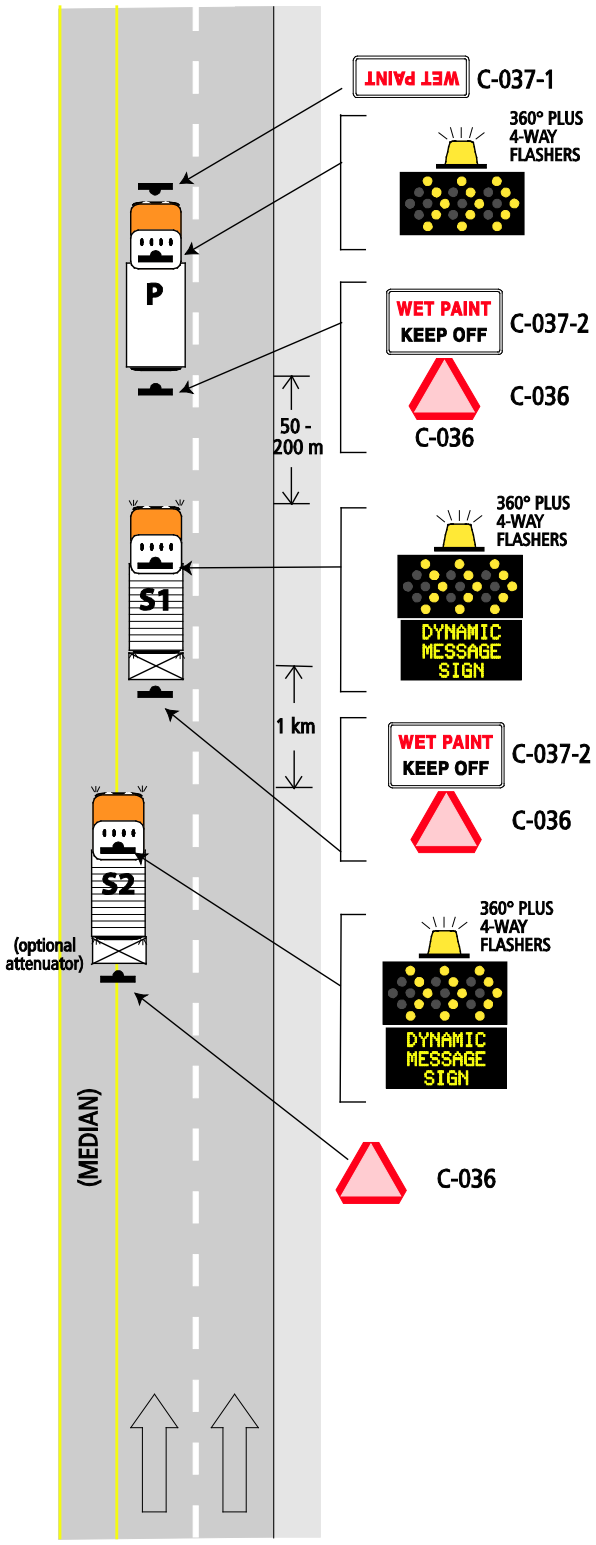
**Guidance:**

- The distance between the Paint Truck and Shadow Vehicle #1 should be determined by the set time (track-free time) of the pavement-marking paint.

**Options:**

- When traffic volumes are high, a third shadow vehicle may be positioned between the Paint Truck and Shadow Vehicle #1. It should follow the Paint Truck at a distance of 50 to 200 metres, followed by Shadow Vehicle #1 at a distance of 200 to 300 metres. The DMS on the third shadow vehicle may display one of these messages:
  - Wet Paint … Keep Right
  - Line Painting … Keep Right
- The third shadow vehicle may be the Escort Vehicle with the DMS repositioned to be rear facing so that the message is visible to rear-approaching traffic.

**Figure 14.4 Quick-Dry Long-Line Marking – Left Lane – Multilane Roadway (Median ≥ 1 m) – Short and Long Duration**



## 14.5 Quick-Dry Long-Line Marking – Left Lane – Multilane Roadway (Median < 1 m, Right Shoulder ≥ 2 m) – Short and Long Duration

**Purpose:**

This layout shows the typical setup for quick-dry long-line marking on the left lane of a multilane roadway when the centre median width < 1 metre wide and the right shoulder is ≥ 2 metres wide.

**Standard:**

**Shadow Vehicle #1 Operations:**

- Shadow Vehicle #1 shall travel in the same lane as the Paint Truck, following at a distance of 50 to 200 metres, depending on sight distance and traffic volumes. (It may have to be closer to the Paint Truck to prevent vehicles from entering the gap between the two vehicles, with consideration given to the amount of set time required for the paint.)
- The DMS messaging may be:
  - right chevron
  - Wet Paint … Keep Right
  - Line Painting … Keep Right

**Shadow Vehicle #2 Operations:**

- Shadow Vehicle #2 shall travel as far right as practicable on the right shoulder, following the Paint Truck at a distance not exceeding 1 kilometre.
- The DMS messaging is specified by the foreman, and may be:
  - Caution mode
  - Line Painting … Ahead
  - Keep Right

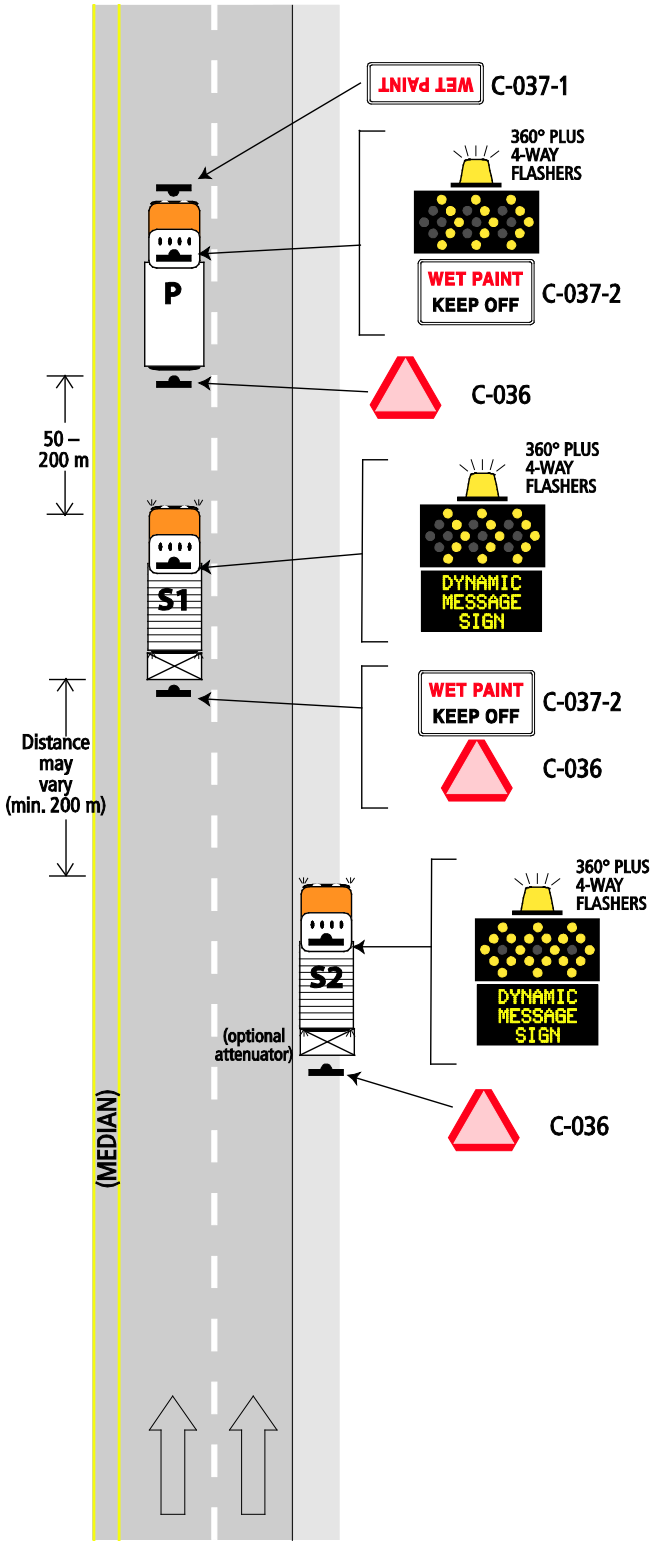
**Guidance:**

- The distance between the Paint Truck and Shadow Vehicle #1 should be determined by the set time (track-free time) of the pavement-marking paint.

**Options:**

- When traffic volumes are high, a third shadow vehicle may be positioned between the Paint Truck and Shadow Vehicle #1. It should follow the Paint Truck at a distance of 50 to 200 metres, followed by Shadow Vehicle #1 at a distance of 200 to 300 metres. The DMS on the third shadow vehicle may display one of these messages:
  - Wet Paint … Keep Right
  - Line Painting … Keep Right
- The third shadow vehicle may be the Escort Vehicle with the DMS repositioned to be rear-facing so that the message is visible to rear-approaching traffic.

**Figure 14.5: Quick-Dry Long-Line Marking – Left Lane – Multilane Roadway (Median < 1 m, Right Shoulder ≥ 2 m) – Short and Long Duration**



## 14.6 Quick-Dry Long-Line Marking – Left Lane – Multilane Roadway (Median < 1 m, Right Shoulder < 2 m) – Short and Long Duration

**Purpose:**

This layout shows the typical setup for quick-dry long-line marking on the left lane of a multilane roadway when the centre median is < 1 metre wide and the right shoulder is < 2 metres wide.

**Standard:**

**Shadow Vehicle #1 Operations:**

- Shadow Vehicle #1 shall travel in the same lane as the Paint Truck, following at a distance of 50 to 200 metres, depending on sight distance and traffic volumes. (It may have to be closer to the Paint Truck to prevent vehicles from entering the gap between the two vehicles, with consideration given to the amount of set time required for the paint.)
- The DMS messaging may be:
  - Right chevron
  - Wet Paint … Keep Right
  - Line Painting … Keep Right

**Shadow Vehicle #2 Operations:**

- Shadow Vehicle #2 shall be outside the travel lane, either in the median or on the shoulder, at a distance not exceeding 1 kilometre.
- The DMS messaging is specified by the foreman, and may be:
  - Right chevron
  - Line Painting … Ahead
  - Keep Right

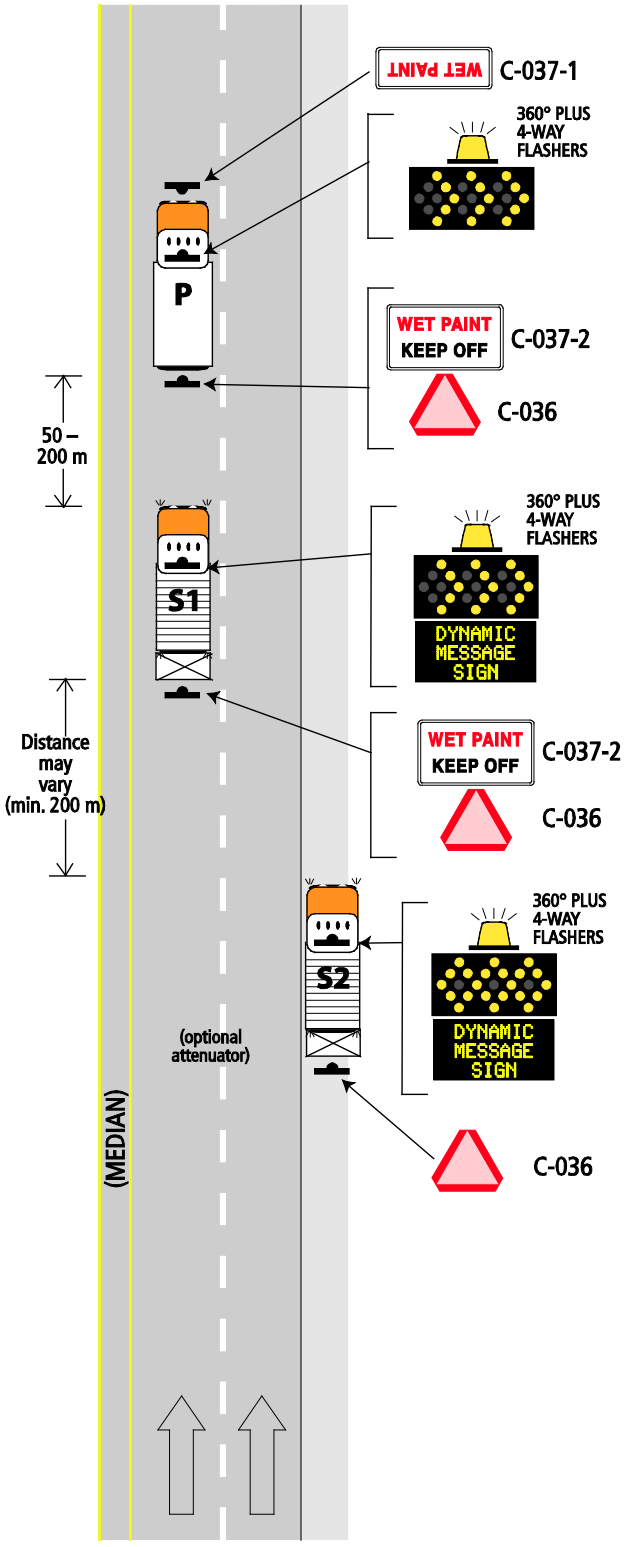
**Guidance:**

- The distance between the Paint Truck and Shadow Vehicle #1 should be determined by the set time (track-free time) of the pavement-marking paint.

**Options:**

- When traffic volumes are high, a third shadow vehicle may be positioned between the Paint Truck and Shadow Vehicle #1. It should follow the Paint Truck at a distance of 50 to 200 metres, followed by Shadow Vehicle #1 at a distance of 200 to 300 metres. The DMS on the third shadow vehicle may display one of these messages:
  - Wet Paint … Keep Right
  - Line Painting … Keep Right
- The third shadow vehicle may be the Escort Vehicle with the DMS repositioned to be rear-facing so that the message is visible to rear-approaching traffic.

**Figure 14.6: Quick-Dry Long-Line Marking – Left Lane – Multilane Roadway (Median < 1 m, Right Shoulder < 2 m) – Short and Long Duration**



## 14.7 Conventional Long-Line Centreline and White Line Marking – Two-Lane, Two-Way Roadway – Short and Long Duration

**Purpose:**

This layout shows typical setups for two kinds of conventional long-line pavement marking along a two-lane, two-way roadway:

- *Figure 14.7 A – Centreline Marking*
- *Figure 14.7 B – White Line Marking (Shoulder or Edge)*

**Standard:**

- Road Marking in Progress C-041-xx signs shall be displayed at each end of the work activity area. The distance between the two signs shall not exceed 10 kilometres.
- The Slow Vehicles Ahead C-045-2A sign shall be positioned after the Road Marking in Progress C-041-xx sign.
- A Caution – Paint Spray Truck Ahead C-043 sign shall be mounted on the front of the Escort Vehicle.
- Both the Paint Truck and the Escort Vehicle shall have:
  - a Pass This Side C-042-LR sign
  - a 360-degree flashing light and 4-way flashers
  - a dynamic message sign (DMS) set to alternate between flashing arrow mode and a displayed message
- Constant communication is required between all Vehicle Operators.
- For centreline marking, cones shall be placed on the centreline.
- For shoulder line marking, cones shall be placed on the white line.

**Guidance:**

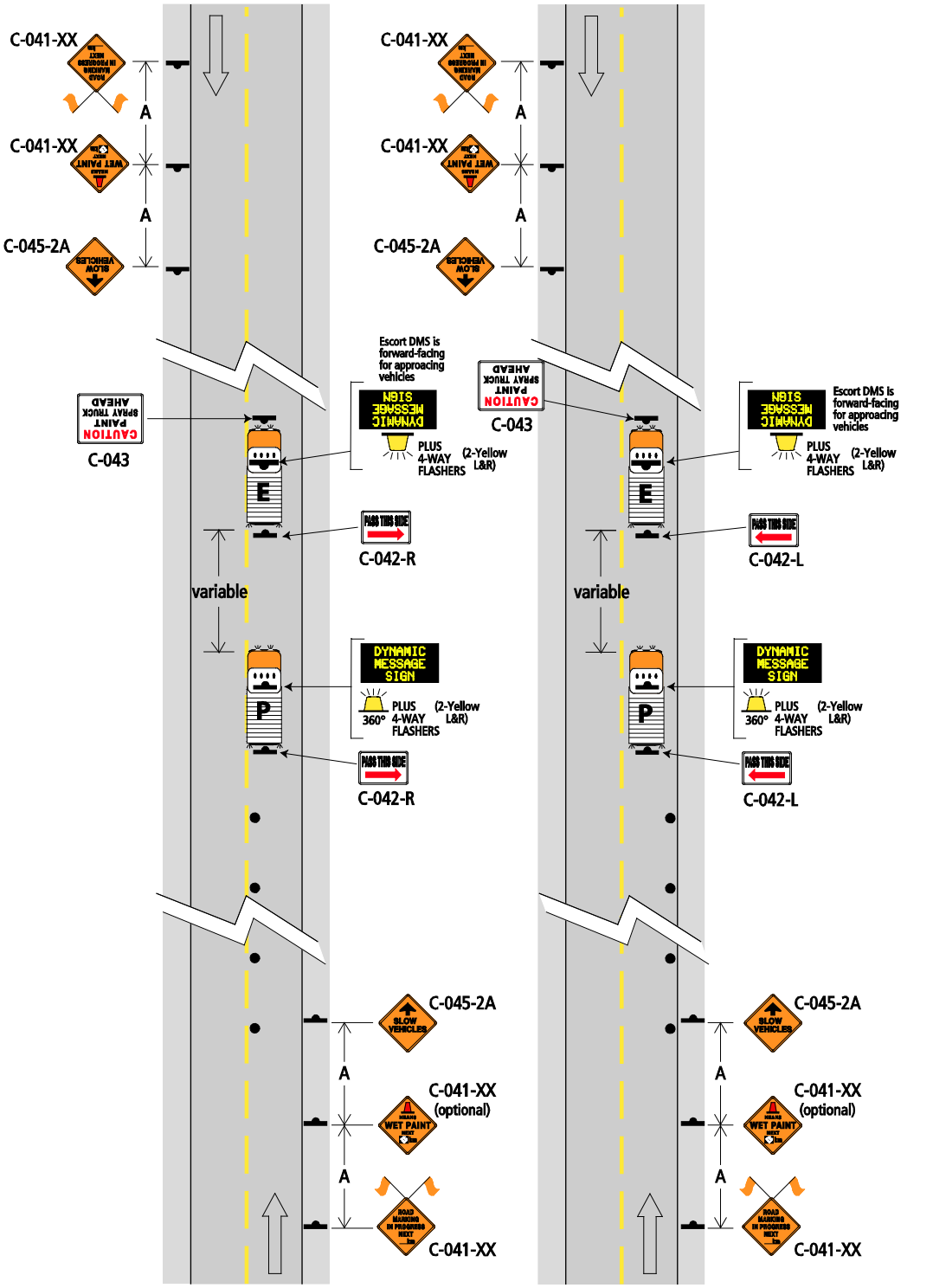
- If it is not practicable for drivers following the Paint Truck and Escort Vehicle to pass the operation, the Paint Truck and Escort Vehicle should pull over periodically to allow these vehicles to go around them.
- The distance between the Escort Vehicle and the Paint Truck will vary, depending on sight lines. They should be as close to each other as possible, but it may be necessary to increase the space on curved roads to provide more advance warning for approaching traffic.

**Options:**

- The Wet Paint side of an additional Road Marking in Progress C-041-xx sign may be used within the work area.
- Crash attenuators may be added to the Paint Truck and/or the Escort Vehicle.
- If shadow vehicles are used to supplement the operation, they shall have dynamic message signs (DMS).

**Figure 14.7: Conventional Long-Line Centreline and White Line Marking – Two-Lane, Two-Way Roadway – Short and Long Duration**

**14.7 A – Centreline Marking** **14.7 B – White Line Marking (Shoulder or Edge)**



## 14.8 Conventional Long-Line Marking – Multilane Roadway – Short and Long Duration

**Purpose:**

This layout shows the typical setup for conventional long-line pavement marking on a multilane roadway.

The diagram shows line painting occurring in different lanes and in different directions—and the associated signing required for each setup—in order to illustrate that the painting operation may occupy either lane. The diagram does not imply a requirement to paint in both directions simultaneously.

**Standard:**

- Road Marking in Progress C-041-xx signs shall be displayed at each end of the work activity area. The distance between the two signs shall not exceed 10 kilometres.
- For short-duration work, a Crew Working Ahead C-004 sign is required.
- For long-duration work, a Construction Ahead C-018-1A sign is required.
- A Left/Right Lane Closed Ahead C-130-L/R sign with a C-130-T distance tab is required in advance of a second Left/Right Lane Closed Ahead C-130-L/R sign.
- Both the Paint Truck and the Shadow Vehicle require two 360-degree flashing lights when painting the centreline or working at night.
- Cones are required along all painted lines—on the centreline for centreline marking and on the white line for shoulder line marking.
- Dynamic message signs (DMS) shall be used on the Paint Truck and the Shadow Vehicle, with the arrow direction indicating the direction of travel and the side on which vehicles are to pass the operation.
- The taper shall be delineated to prevent vehicles from driving in the working lane or pulling in behind the Paint Truck.
- A flashing arrow board (FAB) is required if the speed limit is ≥ 70km/h.
- Constant communication is required between all Vehicle Operators.

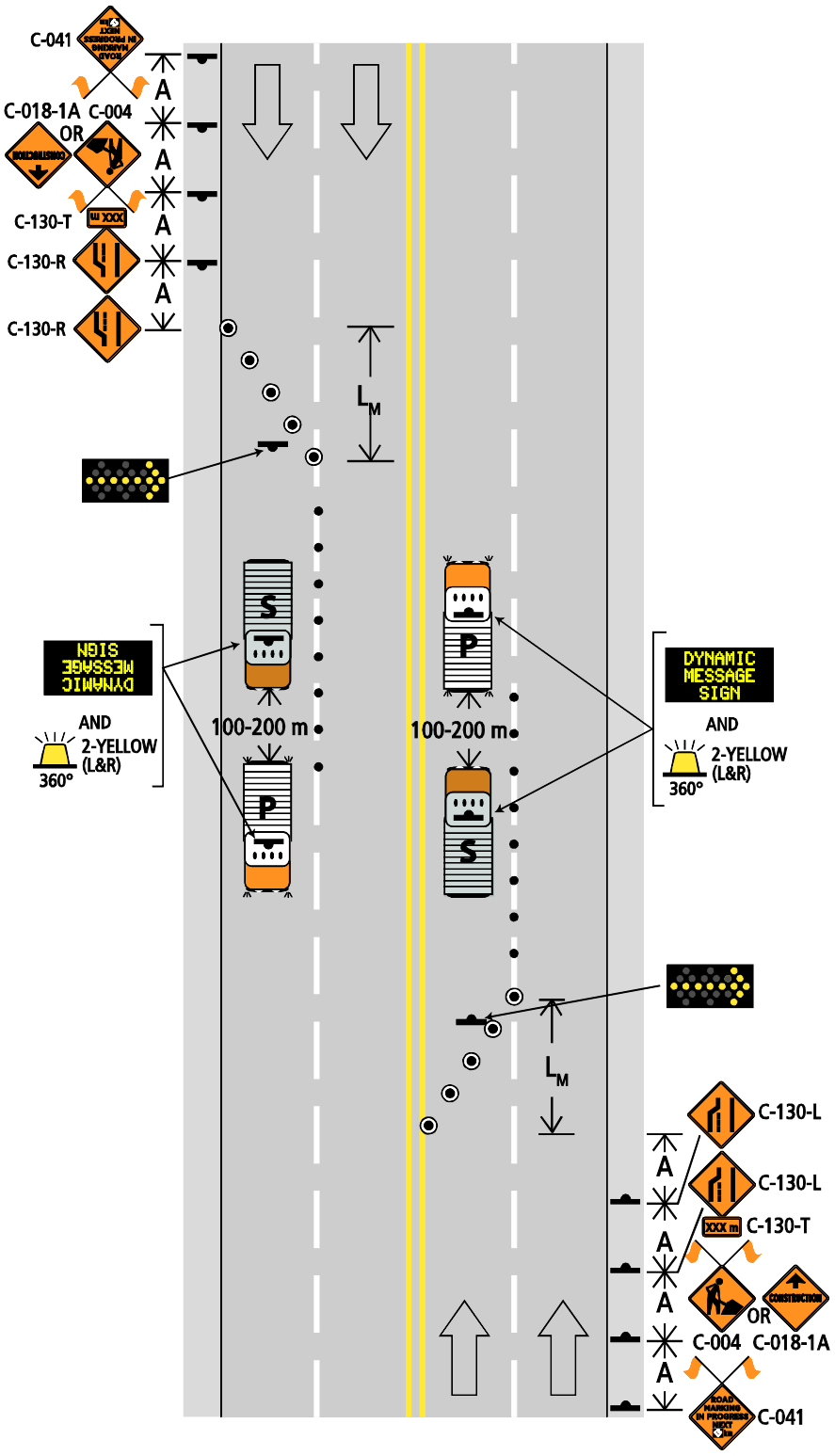
**Guidance:**

- Typical spacing between the Paint Truck and the Shadow Vehicle is 100 to 200 metres.

**Options:**

- If the speed limit is ≤ 60 km/h, a Lane Closure Arrow C-053 sign may replace the flashing arrow board (FAB) in the lane taper.
- The dynamic message signs (DMS) on top of the Paint Truck and Shadow Vehicle may be replaced with flashing arrow boards (FABs).

**Figure 14.8: Conventional Long-Line Marking – Multilane Roadway – Short and Long Duration**



## 14.9 Left-Turn Arrow Marking – Short and Long Duration

**Purpose:**

This layout shows the typical setup for marking left-turn arrows at an intersection.

If the intersection requiring work is not depicted in the diagram, see the layouts in *Section 11: Traffic Control Layouts – Intersections.*

**Standard:**

- Road Marking in Progress C-041-xx signs shall be displayed at each end of the work activity area. The distance between the two signs shall not exceed 10 kilometres.
- For short-duration work, a Crew Working Ahead C-004 sign is required.
- For long-duration work, a Construction Ahead C-018-1A sign is required.
- When the speed limit is ≥ 70 km/h, a vehicle with a flashing arrow board (FAB) or a 360-degree flashing light and 4-way flashers shall be stationed within the upstream island.

**Guidance:**

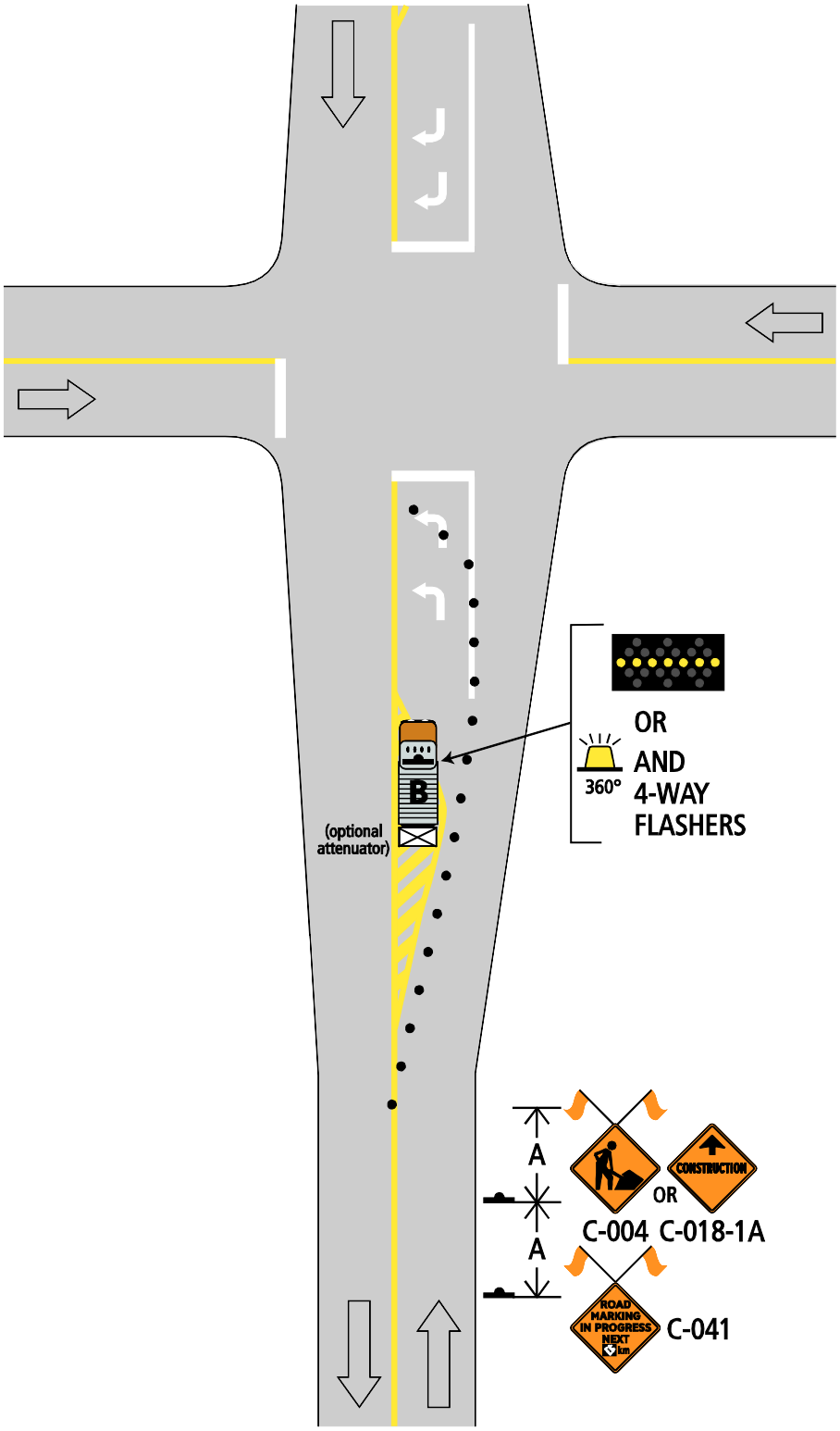
- A work vehicle can be parked on the left-turn loop to activate the left-turn phase of the traffic signal and help to keep vehicles moving in the now-shared through/left-turn lane. If it is parked there for an extended time, the controller may stop activating the left-turn phase so this should be monitored and adjustments made to the setup or traffic control as appropriate.
- If the work allows, keeping a portion of the left-turn slot open near the intersection may provide room for vehicles to make left turns.

**Options:**

- The flashing arrow board (FAB) may be replaced as shown below.

| Speed Limit | Workers or Work Vehicles Present | No Workers or Vehicles Present |
|---|---|---|
| ≤ 60 km/h | 360° flashing light and 4-way flashers. | Barricade and Type A yellow warning light. |
| ≥ 70 km/h | No substitution. | |

**Figure 14.9: Left-Turn Arrow Marking – Short and Long Duration**



## 14.10 Stop Line and Crosswalk Marking – Left Lanes – Multilane Roadway – Short and Long Duration

**Purpose:**

This layout shows the typical setup for marking stop lines and crosswalks in left lanes at an intersection.

For multilane intersections, this is usually done in stages, with the inner lane marking done separately from the outer lane marking to minimize disruption to traffic.

If the intersection requiring work is not depicted in the diagram, see the layouts in *Section 11: Traffic Control Layouts – Intersections.*

**Standard:**

- For short-duration work, a Crew Working Ahead C-004 sign is required.
- For long-duration work, a Construction Ahead C-018-1A sign is required.
- A Left Lane Closed Ahead C-130-L sign with a C-130-T distance tab is required in advance of a second Left Lane Closed Ahead C-130-L sign.
- Flashing arrow boards (FABs) shall be used inside all lane closure tapers.
- If the speed limit is ≥ 70 km/h, a buffer space shall be used.

**Guidance:**

- To maintain traffic flow, it may be necessary to restrict turning movements at the intersection (i.e., prohibit left and/or right turns using Turn Control R-015-L/R signs).
- A buffer vehicle with a 360-degree light and 4-way flashers should be used when workers are on the roadway.
- Depending on the route and traffic volumes, it may be preferable to implement a single lane closure (as opposed to the two closures shown in the diagram).
- When a buffer space is provided, it may be less than the length specified in *Table B – Device Spacing Lengths* if space is limited and the adaptation is approved by the Road Authority.

**Options:**

- A Road Marking in Progress C-041-xx sign may be used in advance of the work.
- For low-volume, low-speed (≤ 60 km/h) roadways:
  - A Lane Closure Arrow C-053 sign may replace the flashing arrow board (FAB).
  - The advance lane drop sign and tab may be removed, and the Crew Working Ahead C-004 sign moved upstream by Table B Distance A.

**Figure 14.10: Stop Line and Crosswalk Marking – Left Lanes – Multilane Roadway – Short and Long Duration**

